

Original Article

The Impact of Generative AI Announcements on Suppliers: Evidence From the Stock Market

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Abstract

The rise of generative AI (GenAI) technology is revolutionizing firm operations and supply chain management, profoundly influencing firms within the GenAI supply chain network. While a few emerging studies have explored the impact of GenAI initiatives on firms' stock performance, it remains unclear how such disruptive technology initiatives affect the stock price movements of their supply chain partners. This paper empirically investigates the impact of firms' GenAI initiatives on the abnormal stock returns of their suppliers by analyzing data on news, supply chain relationships, and daily stock prices. The analysis indicates that, on average, a firm's GenAI announcement leads to a positive abnormal return of 0.27% for its suppliers on the announcement day, with additional evidence of sustained long-term value creation. Heterogeneity analyses reveal that this spillover effect is more pronounced among suppliers with higher R&D intensity, stronger sales growth, closer geographic proximity to the focal firm, and operations in less competitive industries. Furthermore, the positive abnormal returns are significantly greater when the GenAI initiative pertains to product innovation rather than process improvement. Our findings have important implications for supply chain stakeholders navigating the adoption of GenAI technologies.

Keywords

Generative AI, Downstream Innovation, Supplier, Spillover Effect, Abnormal Return, Event Study

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1 Introduction

The rise of generative AI (GenAI), a disruptive information technology (IT), is transforming firm operations and supply chain management due to its significant potential in demand forecasting, inventory management, supply chain optimization, and risk management (Fosso Wamba et al., 2024; Jackson et al., 2024). As the development of disruptive innovation involves a vast ecosystem that includes both upstream and downstream partners

([Kumaraswamy et al., 2018](#)), the broad application of GenAI may have a profound impact on stakeholders within the supply chain.

An emerging stream of research has shown that GenAI can significantly influence a firm's market value, with this effect contingent on firm characteristics. For instance, [Eisfeldt et al. \(2023\)](#) find that firms with higher workforce exposure to GenAI earn higher excess returns in the stock market following the release of ChatGPT. [Bertomeu et al. \(2025\)](#) utilize the ban of ChatGPT in Italy to assess the economic consequences of GenAI and find that the stock market value is linked to factors such as technology exposure, firm size, and firm age. However, the financial implications of GenAI initiatives on supply chain partners and the factors driving this impact have largely been overlooked.

Previous research in Operations Management (OM) has demonstrated positive stock market responses to IT initiatives. Investments in Industry 4.0 technologies and blockchain initiatives, for example, have been shown to result in positive stock market reactions and improved financial performance ([Bai et al., 2022](#); [Klößner et al., 2022](#)). However, the impact of GenAI may differ from these technologies. As a disruptive technology, GenAI has immense potential for productivity enhancement and supply chain management ([Brynjolfsson et al., 2025](#); [Jackson et al., 2024](#)), but it also introduces risks related to data privacy, security, and ethical issues ([Bail, 2024](#); [Fosso Wamba et al., 2024](#)). These unique characteristics suggest that GenAI may influence stock market reactions differently compared to other technologies.

According to the relational perspective of disruptive innovation theory, the impact of disruptive innovations extends beyond individual firms, reshaping existing relationships among industry members and business ecosystems rather than affecting only specific incumbents ([Kumaraswamy et al., 2018](#)). Building on this perspective, GenAI initiatives may generate positive spillover effects on the stock performance of suppliers. When a firm embarks on a GenAI initiative, the associated investment often leads to an increased demand for upstream suppliers, thus raising expectations of their sales. Additionally, investors may view downstream firms' GenAI initiatives as catalysts for suppliers' technology innovation and productivity gains, because downstream IT investments and innovations are known to have positive spillover effects on upstream suppliers ([Cheng and Nault, 2012](#); [Chu et al., 2019](#)).

The relational perspective of disruptive innovation posits that disruptive technologies like GenAI can significantly impact existing roles, relationships, and transactions within a business ecosystem, as they often lead to changes in the underlying system architecture ([Jacobides et al., 2006](#); [Kumaraswamy et al., 2018](#)). Accordingly, the spillover effects of GenAI announcements on suppliers are likely to be contingent on a range of internal and external characteristics. For example, investors may assess suppliers more favorably when they exhibit strong innovation capabilities, particularly in terms of R&D intensity and operation within less competitive industries, two key drivers of innovation ([Baysinger and Hoskisson, 1989](#); [Carter et al., 2021](#)). Suppliers with strong sales growth may also be viewed as better equipped to respond to evolving customer demands, thereby enhancing their compatibility with downstream GenAI innovations ([Fornell et al., 2010](#)). Furthermore, spatial proximity has been shown to facilitate knowledge transfer and innovation spillover, making it a salient factor in determining the magnitude of spillover effects ([Chu et al., 2019](#)).

The strategic orientation of a GenAI initiative, whether it emphasizes product innovation or process improvement, also plays a pivotal role in shaping market perceptions. Product-oriented initiatives typically require more intensive supplier involvement due to changes in

design specifications, the integration of new components, or collaborative development processes (Chang et al., 2022; Eisenhardt and Tabrizi, 1995). In contrast, process-oriented initiatives, which focus on enhancing internal operational efficiencies, generally exert less direct influence on suppliers' technological adaptation. Therefore, the market's response to supplier firms may vary depending on whether the downstream GenAI initiative is directed toward product innovation or process enhancement.

To improve the understanding of the financial implications of GenAI from the perspective of downstream disruptive technological innovation, we aim to address the following questions in this article:

- RQ1. Do GenAI announcements by downstream firms have a positive spillover effect on their suppliers' stock performance?
- RQ2. How do the characteristics of suppliers affect their stock market reactions to GenAI announcements by downstream firms?
- RQ3. Whether product-oriented and process-oriented downstream GenAI announcements have a differential impact on the stock performance of suppliers?

To answer these questions, we identified initial GenAI announcements from firms publicly traded in North America by searching news released from leading agencies. We then conducted an event study to analyze the impact of these announcements on the abnormal stock returns of their suppliers. Our empirical analysis reveals that downstream GenAI initiatives have a positive spillover effect on suppliers' abnormal stock returns, and this spillover effect is stronger under conditions where the supplier shows higher levels of R&D intensity, has higher sales growth, is located closer to the customer, operates within a less competitive industry, or when the announcement focuses on product innovation rather than process improvement. To ensure our findings are not driven by self-selection issues, we employ various identification methods, including propensity score matching, instrumental variable regression, a Heckman sample selection model, and difference-in-differences (DID), for robustness checks.

Our paper makes several important contributions to the literature. First, we contribute to the OM literature by examining the financial spillover effects of GenAI within the supply chain from the perspective of downstream IT innovation. In operations and supply chain management, existing studies regarding information technologies' financial implications primarily focus on the stock performance of firms investing in such technologies (Bai et al., 2022; Klöckner et al., 2022), with almost no research exploring how upstream suppliers respond to their downstream customers' IT initiatives in the stock market. Our research addresses this gap by providing empirical evidence that upstream suppliers' stock market value can be positively affected by downstream customers' disruptive IT innovation, broadening the focus from innovating firms to include supply chain partners.

Second, this study contributes to the OM and information systems literature by examining how factors within and outside suppliers influence their stock market reactions to downstream customers' GenAI initiatives. The existing literature has uncovered various factors that moderate the impact of IT initiatives on stock performance, such as investment type, innovativeness, use case, and industry classification (Bai et al., 2022; Dos Santos et al., 1993; Klöckner et al., 2022). However, these studies predominantly focus on the characteristics of initiating firms or the content of the initiatives, whereas factors influencing the stock performance of supply chain partners are still underexplored. We demonstrate that

suppliers with higher R&D intensity, higher sales growth, closer geographical distance to their customers, those operating in less competitive industries, and those associated with product-oriented downstream innovation experience significantly more positive reactions in the stock market. These findings advance our theoretical understanding of disruptive innovation from a relational perspective by highlighting conditions under which suppliers benefit more from downstream disruptive technologies like GenAI.

Lastly, we contribute to the literature on IT spillover by investigating the impact of GenAI initiatives on suppliers' stock market performance. Existing research on technology spillovers primarily addresses productivity ([Tambe and Hitt, 2014](#)), cost performance ([Atasoy et al., 2018](#)), and innovation ([Byun et al., 2021](#)), with limited attention to market value implications. We reveal that disruptive technologies such as GenAI generate significant spillover effects, reflected in positive abnormal stock returns for upstream suppliers. This result advances our understanding of technology spillovers by illustrating how disruptive innovations reshape industry dynamics and interfirm relationships, extending their influence beyond incumbents and disruptors to include other key stakeholders within the broader ecosystem.

2 Literature Review

2.1 Impact of Advanced Technologies

The OM literature has extensively explored the role of advanced technologies in enhancing supply chain efficiency and performance. Technologies such as IoT, artificial intelligence, and blockchain have demonstrated significant benefits in terms of real-time data sharing, decision-making, and supply chain collaboration ([Choi et al., 2022](#); [Sun and Ji, 2022](#)). Studies have shown that these information technologies lead to improved operational efficiencies, reduced costs, and increased supply chain resilience ([Shen and Sun, 2023](#)). GenAI has gained attention for its ability to generate human-like text and optimize decision-making processes. In operations and supply chain management, GenAI holds the potential for demand forecasting, inventory management, supply chain optimization, and risk management ([Jackson et al., 2024](#)), which benefit operational gains, such as improved customer satisfaction, cost reduction, and process automation ([Fosso Wamba et al., 2024](#)).

Given the significant benefits of advanced technologies, previous studies in OM and finance have documented positive stock reactions to IT initiatives. [Bai et al. \(2022\)](#) find that Industry 4.0 technologies investment announcements lead to positive stock market reactions and improved financial performance, while product-oriented investments and technologies applied across the supply chain have more pronounced stock reactions. Similarly, [Klöckner et al. \(2022\)](#) show that blockchain initiatives result in an average abnormal return (AR) of 0.30% on the announcement day and indicate positive long-term impacts on shareholder value. They further show that the stock market reaction to blockchain announcements is less positive when blockchain is used to trace physical objects, to share sensitive data, or when the blockchain project involves an external IT service provider.

2.2 The Disruptive Nature of GenAI

GenAI differs significantly from traditional advanced technologies due to its ability to autonomously generate high-quality and realistic content, images, and other human-like outputs ([Bail, 2024](#); [Marion et al., 2024](#)). Conventional advanced analytics and machine learning algorithms specialize in numerical and optimization tasks such as predictive

modeling and represent significant progress in artificial intelligence (Chui et al., 2023). However, GenAI marks a step change in its evolution. GenAI is highly versatile and scalable, capable of processing complex inputs and generating human-like outputs without direct human intervention (Eisfeldt et al., 2023). Despite being in its early stages of adoption, GenAI's transformative and disruptive potential is widely recognized (Larsen and Narayan, 2023).

This generative capability can substantially increase labor productivity across various contexts. Brynjolfsson et al. (2025) found that a generative AI-based conversational assistant increased productivity, as measured by issues resolved per hour, by an average of 14%, with the greatest impact observed among novice and low-skilled workers. GenAI can also fundamentally reshape the innovation process and drive product development (Brynjolfsson et al., 2025; Marion et al., 2024). For example, generative AI applications like ChatGPT and Midjourney are uniquely disrupting creative work by generating text, images, and audio from vast datasets and user feedback, significantly impacting jobs that rely on content creation and knowledge-intensive tasks (De Cremer et al., 2023). While earlier technologies often led to gradual improvements in innovation or product design, GenAI has the potential to unlock entirely new dimensions of creativity and innovation (Chui et al., 2023). Additionally, GenAI can play a crucial role in decision-making and collaboration, which were previously considered difficult to automate (Chui et al., 2023). GenAI also shows potential in many operations and supply chain management activities, including demand forecasting, inventory management, quality control, marketing and sales, and supply chain optimization (Chui et al., 2023; Fosso Wamba et al., 2024; Jackson et al., 2024).

The relational perspective of disruptive innovation claims that disruptive innovation will alter the relationship and interaction among the ecosystem members (Kumaraswamy et al., 2018). As a result, GenAI's disruptive capability to generate and innovate sets it apart from other advanced technologies, potentially leading to unique and substantial spillover effects on supplier firms.

2.3 Spillover Effects of Advanced Technologies

The OM literature has studied the spillover effect of a firm's technological innovation on its supply chain partners. Technological spillovers can occur through various mechanisms. Sanders (2005) finds that suppliers investing in IT that aligns with their main buyers or network leaders experience an improvement in their operational performance. Furthermore, Cheng and Nault (2012) find that customers' investments in information technologies promote the productivity of upstream suppliers.

As an advanced and disruptive technology, the specific financial implications of GenAI, especially its spillover effects on supply chain partners, have not been sufficiently explored compared to other IT initiatives. Only a few studies investigate the impact of GenAI on stock market value (Bertomeu et al., 2025; Eisfeldt et al., 2023), with almost no research exploring how stakeholders in the supply chain respond to GenAI initiatives in the stock market. Although some literature has explored the benefits to suppliers from the use of IT in supply chain relationships (Subramani, 2004), these studies are primarily limited to scenarios where suppliers actively participate in joint IT initiatives with their partners. These notable research gaps motivate us to examine our first research question (RQ1).

While extensive literature has uncovered various elements that moderate the impact of IT initiatives on the stock market value of announcing firms, the specific factors influencing

suppliers' stock market responses to their downstream customers' IT initiatives are still not well-defined. The influence of these factors is also likely to differ significantly between suppliers and announcing firms, highlighting the importance of examining the potential contributing factors from the suppliers' perspective. The exploration of these factors leads to our second and third research questions (RQ2 and RQ3).

3 Hypotheses Development

3.1 Overall Impact of GenAI Announcements on Suppliers

Building on prior research that demonstrates how IT initiatives can boost the stock market value of the announcing firms ([Bai et al., 2022](#); [Klöckner et al., 2022](#); [Ranganathan and Brown, 2006](#)), this study adopts the relational perspective of disruptive innovation to argue that investors will also react positively to GenAI initiatives announced by downstream customers in the supply chain.

GenAI announcements send a signal to the stock market, showing that announcing firms highly value innovation in artificial intelligence and are committed to making efforts to utilize GenAI to improve productivity, enhance operational efficiency, or reshape their products and customer service ([Brynjolfsson et al., 2025](#); [De Cremer et al., 2023](#); [Jo, 2023](#)). Beyond indicating their own technological advancement and performance prospects, such announcements can also translate into tangible benefits for their suppliers.

Drawing on the relational perspective of disruptive innovation, which emphasizes how innovation-driven disruptions can reconfigure transactional relationships and value creation processes within an ecosystem ([Kumaraswamy et al., 2018](#)), we identify three key pathways through which GenAI initiatives can affect suppliers.

First, a GenAI initiative by a downstream customer is expected to benefit suppliers who provide products that support the development of the customer's GenAI capabilities ([Jaworski and Kohli, 1993](#)). Some GenAI initiatives can increase demand for complementary technologies and components. For example, when major technology firms announce GenAI initiatives, suppliers such as Nvidia, which provides high-performance GPUs essential for training and running GenAI models, often experience increased demand for their products and subsequently positive market reactions ([Chauhan, 2025](#)).

Second, GenAI adoption by downstream customers can enhance supplier efficiency and productivity by strengthening coordination across inter-organizational systems. A recent example is Walmart's "Trend-to-Product" platform, which uses GenAI to scan global fashion signals and automatically generate mood boards, color palettes, and product concepts. Once a trend is identified, the tool automates early-stage development and produces detailed design packs with manufacturing instructions for suppliers. By shortening the design-to-market cycle from about six months to six to eight weeks, it enables suppliers to adjust production schedules and inventory allocation earlier and more accurately, thereby enhancing their operational efficiency and productivity ([De Shong, 2025](#)). Unlike traditional IT that mainly transmits information, this GenAI application provides more timely and actionable insights, enabling suppliers to respond more quickly and effectively.

Third, downstream GenAI innovation can serve as a catalyst for supplier innovation ([Chu et al., 2019](#)). For example, in the automotive industry, automakers are using GenAI for voice assistants that interpret warnings, suggest fuel options, and answer driver queries. To power these applications, Qualcomm developed its Oryon processor, enhancing computing and AI performance with vehicle sensor integration ([Fried, 2024](#)). In this case, GenAI adoption drives

suppliers to create entirely new products and integration capabilities, going beyond the incremental system upgrades typical of traditional IT.

In summary, the technological innovations of downstream customers can contribute to the demand growth of suppliers and drive them to become more productive and innovative, which are essential for a firm's competitive advantage and growth prospects. Therefore, investors may anticipate these potential benefits for suppliers upon the announcements of GenAI initiatives of downstream customers, leading to a positive reaction in suppliers' stock prices. Thus, we propose the following hypothesis.

H1: GenAI announcements of downstream firms have a positive effect on their suppliers' abnormal stock returns.

3.2 Supplier Characteristics Influencing Stock Market Reactions

3.2.1 Supplier Research and Development (R&D) Intensity. R&D intensity refers to the proportion of a firm's sales that is reinvested into research and development activities, indicating a firm's commitment to innovation and technological advancement ([Baysinger and Hoskisson, 1989](#)). Firms with higher R&D intensity tend to be more innovative, allowing them to adapt more swiftly to rapid technological changes and capture greater market share ([Baysinger and Hoskisson, 1989](#); [Ettlie, 1998](#)). The rise of GenAI is reshaping business offerings ([Jo, 2023](#)), and this technological shift may necessitate upstream suppliers to innovate, redesigning their products and services to meet the evolving needs of their customers ([Chang et al., 2022](#); [Tuli et al., 2007](#)).

Downstream firms' GenAI announcements often indicate that they plan to reconfigure their business, product, and service, thereby influencing their procurement from suppliers. From the relational perspective on disruptive innovation, firms in a business ecosystem must ensure that their components remain compatible as disruptive technologies reshape the industry ([Kumaraswamy et al., 2018](#)). As such, suppliers with higher R&D intensity may be perceived as more capable of understanding and adapting to changes in customer needs. They are also more likely to be technologically aligned with downstream firms due to their inherent focus on research and development. Consequently, the market may view these suppliers as more adaptable and capable of maintaining their competitive edge in the wake of their downstream customers' GenAI initiatives, resulting in a favorable abnormal stock return. Thus, we propose the following hypothesis.

H2: The abnormal stock return is higher for suppliers with a higher level of R&D intensity.

3.2.2 Supplier Sales Growth. The abnormal stock returns due to downstream customers' GenAI initiatives may be stronger for suppliers with higher sales growth. The GenAI announcement by a downstream customer signals a disruptive technological innovation within the firm, indicating that the products or services provided by suppliers may need to undergo significant technological change, as it is crucial to integrate relevant customer dynamics into a firm's product innovation process ([Christensen and Bower, 1996](#)) and to ensure that technology modules or product platforms are aligned within the evolving ecosystem ([Kumaraswamy et al., 2018](#)). Thus, suppliers need to intensify their investments or allocate resources to develop new products or processes driven by downstream GenAI innovations to meet customers' demands ([Christensen and Bower, 1996](#)). We posit that suppliers with high sales growth are more likely to be viewed by investors as benefiting significantly from their customers' technological advancements.

On the one hand, high sales growth is a sign of healthy financial performance. Investors may anticipate that these suppliers are more capable of investing in technological innovation to quickly align their offerings with the emerging needs of their customers. On the other hand, suppliers with high sales growth may understand what the customer needs better (Fornell et al., 2010) and focus more on customer-centric innovation (Selden and MacMillan, 2006). Therefore, these suppliers are better positioned to adjust their products and services to align with customers' desires. Given the advantages possessed by high-growth suppliers, we propose the following hypothesis.

H3: The abnormal stock return is higher for suppliers with a higher level of sales growth.

3.2.3 Supplier–Customer Distance. The stock price reaction of suppliers to downstream GenAI announcements may also be influenced by the geographical distance between them, with the effect being stronger when the distance is shorter. Although downstream disruptive innovation may drive suppliers to engage in collaborative innovation to stay aligned with the disruption, the innovation outcome depends on the nature of the partners involved (Faems et al., 2005; Kumaraswamy et al., 2018). Supplier–customer geographic proximity may promote supplier innovation because it is conducive to timely feedback (Chu et al., 2019). Consequently, suppliers closer to their customers can engage in innovation activities more efficiently to address the technological change brought about by customers' adoption of GenAI technologies. Moreover, geographical proximity between suppliers and customers enhances logistical efficiency, reducing transportation costs and time (Chu et al., 2019; Müller et al., 2023), which can foster a more agile supply chain (Müller et al., 2023). This agility helps suppliers swiftly respond to shifts in customer demand spurred by GenAI and achieve a higher performance (Swafford et al., 2008). Overall, closer geographical distance between suppliers and customers announcing GenAI initiatives is beneficial for innovation spillover and supplier performance, thereby amplifying market optimism towards these suppliers. Thus, we propose the following hypothesis.

H4: The abnormal stock return is higher for suppliers whose geographical distances with the announcing firms are smaller.

3.2.4 Supplier Industry Competitiveness. A supplier's stock market reaction to downstream GenAI initiatives may be influenced by the level of industry competitiveness, which plays a critical role in shaping innovation incentives. The introduction of GenAI can drive shifts in market demand, as disruptive technologies often alter customer preferences and redefine value propositions (Henderson, 2006). In response, suppliers must innovate to align with these evolving expectations. In highly competitive industries, where demand is more elastic, firms face stronger pressure to differentiate through quality and product innovation in order to maintain or grow market share (Carter et al., 2021; Raith, 2003). Consequently, suppliers operating in such environments are more likely to prioritize innovation in response to GenAI innovations. Furthermore, industry competition may amplify the magnitude of IT spillovers from downstream firms. For example, Cheng and Nault (2012) show that lower industry concentration on the supplier side relative to customer industry concentration, implying higher competition, can increase the IT spillover effects and improve supplier productivity. These benefits, including increased innovation and productivity, suggest that suppliers in highly competitive industries are better positioned to capitalize on downstream GenAI initiatives. As a result, investors may perceive these suppliers as more likely to derive value from such technological disruptions, leading to more favorable stock market reactions.

Conversely, it is conceivable that the market regards suppliers in less competitive industries as being better positioned to capitalize on the benefits of customers' GenAI initiatives compared to those in highly competitive industries. GenAI significantly impacts businesses by transforming products, services, and operational processes (Brynjolfsson et al., 2025; De Cremer et al., 2023; Jo, 2023). During the technological transformation of downstream customers, suppliers within highly competitive industries may face greater challenges than those in less competitive sectors. The potential demand shifts resulting from the customer's GenAI initiatives can pose an innovation risk for suppliers, as they may struggle to maintain adequate levels of innovation to meet the customer's needs (Carter et al., 2021). In this case, suppliers have to quickly update their offerings to align with their customers' technological innovation to maintain customer satisfaction (Christensen and Bower, 1996; Sanders, 2005). Failure to do so may erode their market position due to the intense competition. Based on these arguments, we propose two competing hypotheses below:

H5a: The abnormal stock return is higher for suppliers situated in a more competitive industry.

H5b: The abnormal stock return is higher for suppliers situated in a less competitive industry.

3.3 Product– Versus Process–Oriented GenAI Announcements

The strategic focus of a downstream firm's GenAI initiative can influence the market's perception of its impact on suppliers. Product-oriented GenAI announcements, such as those involving the development of AI-enhanced features or entirely new AI-powered products, often signal a heightened need for supplier adaptation and collaboration, particularly given the disruptive nature of GenAI technologies. These initiatives may require suppliers to develop new components, integrate advanced technology modules, or engage in co-development activities (Eisenhardt and Tabrizi, 1995; Kumaraswamy et al., 2018). In contrast, process-oriented GenAI initiatives, which aim to improve internal productivity, streamline supply chain operations, or enhance customer service, primarily focus on operational efficiencies and typically impose fewer direct technological demands on suppliers.

Product-oriented initiatives, therefore, are more likely to generate innovation spillovers and foster long-term collaboration between downstream firms and their suppliers. In comparison, process-oriented initiatives are generally associated with internal improvements that may not directly benefit upstream partners. As a result, investors may interpret product-oriented GenAI announcements as stronger signals of future growth potential for suppliers. This leads to the following hypothesis:

H6: The abnormal stock return is higher for suppliers when the GenAI announcement is product-oriented rather than process-oriented.

4 Data and Model

4.1 Data

For the purpose of this study, we collected three types of data: firms with GenAI announcements, suppliers of announcing firms, and stock market returns of suppliers.

4.1.1 Firms With GenAI Announcements. To identify firms with GenAI-related announcements, we leveraged the LexisNexis database to search the leading news agencies, including PR Newswire, Business Wire, and GlobeNewswire, from 1 January 2023 to 31 May 2024. Our search strategy focused on all announcements related to GenAI within each firm's public disclosures. The search employed keywords such as AIGC, Generative AI, GenAI, ChatGPT, large language models, generative artificial intelligence, NLP, and related terms. After removing duplicated articles referring to the same GenAI initiatives, our initial search yielded a total of 14,941 potential announcements. Subsequently, we screened all articles based on the following criteria:

- (1) We focused on 2,084 announcements from firms listed in the North American market to maintain consistency in market behaviors and regulatory conditions, thereby ensuring a more reliable analysis.
- (2) We manually reviewed and verified each announcement to determine its relevance to GenAI. Only announcements that explicitly referenced specific GenAI initiatives were retained. These included instances involving investments in GenAI, the adoption of GenAI tools, the integration of GenAI technologies into products, or the incorporation of GenAI into organizational workflows. Examples of qualifying announcements are provided in [Appendix EC.1](#). A total of 1,397 announcements that only broadly reference “GenAI and other information technologies” were excluded.
- (3) As investors might value multiple announcements from one firm differently ([Klößner et al., 2022](#)), we retained only the earliest GenAI initiative for each company within the sample period.
- (4) We excluded 36 firms with a stock price below 1 USD on the announcement day to mitigate the effects of using penny stocks ([Konchitchki, 2011](#)).

Following the above criteria, we identified 254 GenAI announcements each from a distinct firm. Within these announcements, firms employ GenAI for product manufacturing, service enhancement, efficiency improvement, and related areas. As accurate announcement dates are essential for the validity of this study, we manually verified the announcement release dates. We moved announcements made on non-trading days, such as weekends or public holidays, to the next trading day. If an announcement occurred after the stock market closed, we recorded the announcement date as the following trading day.

4.1.2 Suppliers of Announcing Firms. We employed the Compustat Segment database to extract the suppliers of the announcing firms, restricting our search of suppliers to listed companies in the North American market. To ensure relevance, the extraction of supplier–customer relationships was confined to the five years preceding the announcing year ([Hertzel et al., 2008](#)). We further searched the Factset database, another widely used resource for identifying a firm's disclosed suppliers ([Chen et al., 2023](#)), to match as many suppliers as possible. This process resulted in a total of 657 suppliers of 178 announcing firms.

Given that this study focuses on the spillover effects of downstream customers' GenAI initiatives, we excluded supplier–customer relationships where the supplier itself made a GenAI announcement before the customer's announcement date. This was done to avoid the confounding impact of the supplier's own GenAI announcement. Since different announcing firms may share the same supplier, some suppliers are associated with more than one announcement date. To eliminate potential interference among multiple announcements of

different customers, we excluded suppliers with more than one announcing firm if these announcing firms have overlapping announcement dates. To mitigate the effects of confounding events, we further excluded suppliers with confounding events (i.e., board change, earnings, and merge/acquisition) within a four-day window around the GenAI announcements of their customers (i.e., days -2 , -1 , 0 , and 1), as adopted by [Lo et al. \(2018\)](#). This screening criteria left us with 580 suppliers of 133 announcing firms. Finally, we excluded suppliers if the estimation period for the event study was under 200 days, their stock price dropped below 1 USD during this period, or they had missing variables. Since the same supplier can be linked to multiple announcing firms across various announcements, our final sample includes 277 suppliers of 117 announcing firms, resulting in 515 unique supplier–announcement observations.

[Table A2](#) in [Appendix EC.2](#) summarizes the steps for constructing our sample. Each row reports an additional data requirement, along with the resulting numbers of GenAI announcements and suppliers. The summary statistics of the supplier sample, including the distribution across industry and country, are presented in [Table A3](#) in [Appendix EC.3](#).

4.1.3 Financial Data. For calculating daily ARs, we obtained daily stock prices of all suppliers from Compustat. Given that our sample firms are based on the North American market, we employ the S&P 500 index as the stock market index while calculating ARs. North American Fama–French factors were retrieved from the Dartmouth College database.

4.2 Variables

4.2.1 Independent Variables. *Supplier R&D intensity.* We employ the ratio of R&D expenditure to sales in the fiscal year prior to the announcing year as a measure of firm R&D intensity, which is the most widely used metric to capture the investment intensity of a firm in innovation activities ([Xue et al., 2012](#)).

Supplier Sales Growth. We measure sales growth using the firm's sales growth rate in the fiscal year prior to the announcement. For example, a firm's sales growth in 2022 is calculated by the increase in sales from 2021 to 2022, divided by the sales in 2021.

Supplier–Customer Distance. To calculate the geographical distance between suppliers and announcing firms, we first obtained zip codes of both parties' headquarters from Compustat, and then used zip codes to retrieve the latitude and longitude coordinates of their headquarters ([Akturk and Ketzenberg, 2022](#); [Reuer and Lahiri, 2014](#)). We calculate the geographical distance between suppliers and announcing firms using the great circle distance formula ([Bode and Wagner, 2015](#); [Chu et al., 2019](#)), based on an Earth radius of 3,963 miles. The great circle distance formula is given by $3,963 \times \arccos[\sin(Lat_1) \times \sin(Lat_2) + \cos(Lat_1) \times \cos(Lat_2) \times \cos(Long_2 - Long_1)]$, where Lat_1 and Lat_2 are the latitudes, and $Long_1$ and $Long_2$ are the longitudes, of two points, respectively. Considering that the distribution of distance is right skewed, we log transform the distance after adding one ([Chu et al., 2019](#)).

Supplier Industry Competitiveness. We use industry concentration as a proxy for the level of competition within a three-digit NAICS industry. We obtained firm concentration data for the three-digit NAICS industries from the 2017 Economic Census Data on the U.S. Census Bureau website. Given that the Herfindahl–Hirschman index (HHI) is limited to manufacturing industries, and our sample includes non-manufacturing firms, we adopt the market share of

the top 20 firms within each three-digit NAICS industry as the measurement of industry concentration ([Cheng and Nault, 2012](#)).

Product vs. Process-Oriented Announcement. To assess the differential impact of various GenAI initiatives on supplier stock returns, we manually categorize the GenAI announcements based on whether they focus on product innovation or process improvement. Specifically, we classify announcements as product-oriented if they mention GenAI-enhanced product features, new GenAI-powered products, or innovations where GenAI is directly integrated into the firm's offerings. In contrast, process-oriented announcements involve the application of GenAI in internal operations, such as supply chain optimization, customer service automation, and workflow efficiency enhancements.

4.2.2 Control Variables. Following the existing literature, we incorporate several firm-level control variables into our analyses: firm size, firm age, market-to-book ratio (MTB), leverage, and relationship duration. Individual events are typically less likely to result in significant stock price movements for larger firms ([Klöckner et al., 2022](#)), while older firms usually have more resources and experience in technological innovation. MTB represents the potential for firm growth and may influence the reaction of investors ([Klöckner et al., 2022](#)). Leverage indicates the firm's capital structure and could influence financial risk ([Hendricks et al., 2015](#)). Relationship duration has been used as a proxy for the strength, closeness, and coordination efficiency between suppliers and announcing firms ([Song et al., 2023](#)), which may influence suppliers' stock market reactions. To account for potential confounding effects stemming from industry-wide GenAI adoption, we construct an industry-level control variable defined as the logarithm of the number of monthly GenAI announcements within each supplier's two-digit NAICS industry plus one. Additionally, we include a binary variable (*Same Industry*) indicating whether suppliers and announcing firms operate within the same two-digit NAICS industry, capturing potential intra-industry spillovers. [Table 1](#) provides definitions and descriptive statistics for key variables, and their correlation matrix is presented in [Appendix EC.3](#).

Table 1. Variable definitions and descriptive statistics.

Variables	Definition	Obs	Mean	SD	Min	Max
AR ₀	The abnormal stock return of a supplier on the announcement day 0	515	0	0.02	−0.05	0.06
Supplier R&D Intensity	The ratio of R&D expenditure to sales in the fiscal year prior to the announcement	515	0.10	0.12	0	0.77
Supplier Sales Growth	Percentage change in sales in the fiscal years prior to the announcement	515	0.10	0.19	−0.34	1.02
Supplier–Customer Distance	The logarithm of the geographical distance between the supplier and the announcing firm plus one	515	6.47	1.96	0	9.07
Supplier Industry Concentration	Market share of the largest 20 firms within each three-digit NAICS industry	515	0.38	0.18	0.11	0.86
Product-Oriented Announcement	A binary variable set to 1 if the announcement focuses on product innovation, and 0 otherwise	515	0.84	0.37	0	1
Size	The logarithm of the total assets in the fiscal year prior to the announcement	515	9.84	2.34	4.09	13.04
Age	The number of years since the first appeared in Compustat	515	31.37	19.29	4	74
MTB	Ratio of stock closing price to book value per share for the announcement date	515	14.37	42.30	−16.62	284.02
Leverage	Ratio of total liabilities to total assets in the fiscal year prior to the announcement	515	0.58	0.24	0.14	1.10
Relationship Duration	The number of years since a supplier was first recorded as the announcing firm's supplier in the Factset or Compustat database	515	4.46	4.06	1	21
Announcement Intensity	The logarithm of the monthly number of GenAI announcements in each two-digit NAICS industry of the supplier, plus one	515	1.73	1.31	0	4.08
Same Industry	A binary variable set to 1 if the supplier and the announcing firm are located in the same two-digit NAICS industry, and 0 otherwise	515	0.25	0.43	0	1

Notes: The data are summarized at the supplier-announcement level. AR = abnormal return; R&D = Research and Development; MTB = market-to-book ratio; GenAI = generative artificial intelligence.

4.3 Model

Following the standard practice in the literature, we use the event study method to quantify the impact of GenAI announcements on the abnormal stock returns of suppliers. This method examines the financial implications of an event by computing the ARs associated with that event in a narrow event window (Dewan and Ren, 2007; Pu et al., 2025). A key advantage of this approach is its ability to isolate the effects of focal events from confounding events that are fairly common in an extended window, such as product launches, executive changes, earnings announcements, and supply chain disruptions. Our empirical analysis primarily consists of two parts. In the first part, we use the classical event study approach to assess the overall impact of downstream GenAI announcements on the abnormal stock returns of suppliers. In the second part, we utilize the extended event study approach with covariates (Bai et al., 2022; Klöckner et al., 2022), operationalized as a cross-sectional regression model, to examine how this impact is influenced by the characteristics of suppliers and the announcement focus.

4.3.1 Overall Impact of GenAI Announcements on Suppliers. To operationalize this analysis, we converted calendar days into trading days, where day 0 represents the announcement date, day −1 denotes the preceding trading day, and day 1 signifies the trading day following the event (Klöckner et al., 2022). If the announcement was made on a weekend, holiday, or after market hours, the next trading day was considered the announcement date. ARs are computed as the difference between observed firm returns and expected returns. We use the

Fama–French four–factor model to estimate expected returns (Carhart, 1997; Fama and French, 1993):

$$R_{it} = \alpha_i + \beta_i R_{mt} + \gamma_i SMB_{it} + \delta_i HML_{it} + \theta_i UMD_{it} + \epsilon_{it}, \quad (1)$$

where R_{it} represents the observed return for stock i on day t , R_{mt} is the stock market return of the S&P 500 index on day t , SMB_t is the small–minus–big–size portfolio return on day t , HML_t is the high–minus–low book–to–market portfolio return on day t , and UMD_t is the momentum factor. The linear regression coefficients are denoted as α_i , β_i , γ_i , δ_i , and θ_i , and ϵ_{it} represents the error term. Following the event study literature, these coefficients were estimated from a 200 trading day period spanning from day -210 to day -11 . The 10 trading days prior to the announcement date were excluded to rule out the potential confounding impact of any other events immediately preceding the GenAI announcements. Then, we use the estimates $\hat{\alpha}_i$, $\hat{\beta}_i$, $\hat{\gamma}_i$, $\hat{\delta}_i$, and $\hat{\theta}_i$ to calculate AR that represent the disparity between the observed return and the expected return:

$$AR_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{mt} + \hat{\gamma}_i SMB_t + \hat{\delta}_i HML_t + \hat{\theta}_i UMD_t). \quad (2)$$

Consistent with past event studies, we consider the ARs from day -1 to day 1 (Lo et al., 2018). We use a t -test to examine whether the average AR of suppliers on the announcement day is significantly different from zero, allowing us to test H1. Additionally, we employ a Wilcoxon signed–rank test to investigate whether the median AR of suppliers is significantly different from zero and a binomial sign test to assess whether the positive rate of ARs is significantly $> 50\%$.

4.3.2 Characteristics Influencing Stock Market Reactions. We use the following cross–sectional regression model to investigate how the impact of downstream GenAI announcements on suppliers' ARs is affected by the characteristics of suppliers and the announcement focus:

$$\begin{aligned} AR_{i0} = & \beta_0 + \beta_1 R\&D_i + \beta_2 Sale\ Growth_i + \beta_3 Geo\ Distance_i \\ & + \beta_4 Industry\ Concentration_i \\ & + \beta_5 Product\ Oriented\ Announcement_i \\ & + \beta_6 Size_i + \beta_7 Age_i + \beta_8 MTB_i \\ & + \beta_9 Leverage_i + \beta_{10} Rel\ Duration_i \\ & + \beta_{11} Announcement\ Intensity_i \\ & + \beta_{12} Same\ Industry_i + \epsilon_i, \end{aligned} \quad (3)$$

where AR_{i0} refers to the abnormal stock return of supplier i on the announcement day 0, as computed using the Fama–French four–factor model. We use the AR on the announcement day as the dependent variable to capture the immediate market reaction to GenAI initiatives, reflecting investors' revised expectations about supplier value, consistent with the approach of Bai et al. (2022) and Klöckner et al. (2022). β_1 , β_2 , β_3 , β_4 , and β_5 denote the coefficients of the independent variables *Supplier R&D Intensity*, *Supplier Sales Growth*, *Supplier–Customer Distance*, *Supplier Industry Concentration*, and *Product–Oriented Announcement*,

respectively. Testing whether these coefficients are significantly different from zero allows us to evaluate Hypotheses H2–H6. We further impose fixed effects at the announcing firm level and the supplier industry level when applicable. Because only the first GenAI announcement of each firm is retained, the announcing–firm fixed effect also captures the influence of announcement timing. All continuous variables are winsorized at their first and 99th percentiles to limit the impact of outliers.

5 Empirical Results

In this section, we begin by presenting the results of the event study without covariates to examine the overall impact of downstream GenAI announcements on supplier stock performance. Next, we extend the analysis by incorporating covariates to examine how supplier characteristics and the strategic focus of GenAI initiatives influence suppliers' ARs. Finally, we perform a series of robustness checks to assess the consistency and validity of our results.

5.1 Overall Impact of GenAI Announcements on Suppliers

Table 2 reports the results of statistical tests conducted on the average AR, median AR, and the proportion of positive ARs from day -1 to day $+1$, where day 0 denotes the announcement date of GenAI initiatives by downstream firms. Specifically, a t -test is employed to assess whether the average AR across suppliers significantly deviates from zero; the Wilcoxon signed–rank test evaluates whether the median AR differs significantly from zero; and the binomial sign test examines whether the proportion of positive ARs significantly deviates from 50%. These methods are commonly applied in event studies within the OM literature (Bai et al., 2022; Hendricks et al., 2015; Klöckner et al., 2022).

Table 2. Event study results on suppliers' stock market reaction.

	Day -1 (1)	Day 0 (2)	Day 1 (3)
Average abnormal return	−0.0008	0.0027***	0.0012
t -test (t -statistic)	(−1.0061)	(3.4786)	(1.4530)
Median abnormal return	−0.0007	0.0013***	0.0013*
Wilcoxon signed rank test (Z -statistic)	(−0.9362)	(2.7806)	(1.8664)
Positive rate of the abnormal returns	0.4699	0.5456**	0.532
Binomial sign test (Z -statistic)	(−1.3660)	(2.0711)	(1.4542)
Observations	515	515	515

Notes: All tests are two-tailed. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. We provide additional information on the industry distribution and descriptive statistics of suppliers with positive versus non-positive abnormal returns in Appendix EC.4. When separating generative artificial intelligence (GenAI) announcements by orientation, we find that product-oriented announcements are associated with significant positive abnormal returns for suppliers, whereas process-oriented announcements have no statistically significant effect (Appendix EC.5).

Column 1 shows no significant stock market reaction on the day before the GenAI announcement. However, Column 2 reveals a significant positive abnormal stock return on the day of the announcement. Specifically, the two–tailed t -test indicates that the mean AR of 0.27% is significantly > 0 at the 1% significance level. Moreover, the median positive abnormal stock return (0.13%) is also significantly positive at the 1% significance level. Additionally, the percentage of positive ARs is 54.56%, which is significantly different from 50% at the 5% significance level. Moving to Column 3, the mean AR on day 1 is 0.12%, and does not differ significantly from zero. The median positive abnormal stock return (0.13%) for

day 1 differs significantly from zero at the 10% level, and the percentage of positive AR (53.2%) does not differ significantly from 50%. The diminished impact on day 1 is expected as the stock market is generally highly efficient in absorbing price signals. These findings offer support for H1, indicating that the abnormal stock returns of suppliers are indeed driven by the GenAI announcements of downstream customers.

5.2 Characteristics Influencing Stock Market Reactions

To test Hypotheses 2 through 6, we perform cross-sectional regression analyses using three slightly different model specifications. The results are reported in [Table 3](#). Columns 1 and 2 include fixed effects for the announcing firm, which prevents the estimation of the *Product-Oriented Announcement* coefficient. Furthermore, Columns 2 and 3 incorporate fixed effects at the supplier industry level, limiting the identification of *Supplier Industry Concentration* due to insufficient time-varying variation in this variable. Accordingly, we focus on Column 1 when evaluating H5a and H5b, and on Column 3 when assessing H6.

Table 3. Cross-sectional regression results on factors influencing stock market reactions.

Variables	Abnormal return on day 0 (AR_0)		
	(1)	(2)	(3)
<i>Supplier R&D Intensity</i>	0.0221** (0.0088)	0.0232*** (0.0087)	0.0259*** (0.0058)
<i>Supplier Sales Growth</i>	0.0148*** (0.0049)	0.0141** (0.0056)	0.0115** (0.0049)
<i>Supplier-Customer Distance</i>	-0.0010** (0.0004)	-0.0011** (0.0005)	-0.0011*** (0.0004)
<i>Supplier Industry Concentration</i>	0.0122** (0.0056)	0.0011 (0.0071)	0.0007 (0.0053)
<i>Product-Oriented Announcement</i>			0.0046** (0.0021)
<i>Size</i>	-0.0007* (0.0004)	-0.0005 (0.0004)	-0.0009*** (0.0003)
<i>Age</i>	-0.0000 (0.0001)	-0.0000 (0.0001)	0.0000 (0.0001)
<i>MTB</i>	0.0000 (0.00002)	0.0000 (0.00002)	0.0000 (0.00002)
<i>Leverage</i>	0.0034 (0.0035)	0.0047 (0.0036)	0.0026 (0.0033)
<i>Rel Duration</i>	-0.0003 (0.0003)	-0.0004 (0.0003)	-0.0003* (0.0002)
<i>Announcement Intensity</i>	0.0008 (0.0009)	0.0015 (0.0030)	-0.0001 (0.0013)
<i>Same Industry</i>	-0.0040 (0.0034)	-0.0044 (0.0041)	-0.0005 (0.0021)
Announcing firm FE	Yes	Yes	No
Industry FE	No	Yes	Yes
Observations	515	515	515
R^2	0.316	0.338	0.128

Notes: The robust standard errors reported in parentheses are clustered at announcing firm level. The results are similar when the standard errors are clustered at the supplier level (Appendix EC.6). All continuous variables are winsorized at their first and 99th percentiles. R&D = Research and Development; MTB = market-to-book ratio; FE = fixed effect. Significance at $*p < 0.1$; $**p < 0.05$; $***p < 0.01$.

Overall, the results show that suppliers with greater R&D intensity and higher sales growth experience a more positive stock market reaction to their customers' GenAI announcements, supporting H2 and H3, respectively. Furthermore, in support of H4, the geographical distance between the supplier and the announcing firm significantly attenuates this positive stock market reaction. The effect of *Supplier Industry Concentration* in Column 1 indicates that firms operating in industries with higher concentration exhibit significantly higher abnormal stock returns, suggesting that industry competitiveness mitigates the positive stock market reaction of the supplier, which supports H5b instead of H5a. Additionally, the coefficient of *Product-Oriented Announcement* in Column 3 indicates that product-oriented GenAI initiatives generate significantly higher abnormal stock returns than their process-oriented counterparts, thereby supporting H6.

5.3 Robustness Checks

5.3.1 Robustness of Event Study to Self-Selection. One potential concern with our event study in Section 5.1 is that it examines a selective sample of suppliers whose downstream customers have made GenAI announcements. Given that these customers tend to be highly strategic when choosing their suppliers (Xue et al., 2018), the positive ARs observed in our sample might stem from the unique observable and/or unobservable characteristics of these suppliers rather than from technology spillover within the supply chain. We address this issue using both propensity score matching (PSM) and instrumental variable regression.

Propensity Score Matching. Since our supplier sample is relatively small, a potential concern is that our findings might be driven by the specific composition of the supplier sample, which might represent a set of firms that have experienced positive ARs due to certain broader factors unrelated to GenAI announcements (e.g., recovery from prior supply chain disruptions or favorable regulatory changes). To address this concern, we use PSM to identify a set of suppliers with similar observable characteristics to those in our sample, but whose downstream customers have not made GenAI announcements. If the ARs are indeed driven by the GenAI announcements, we would expect the matched suppliers, who did not experience GenAI announcements, to show no significant ARs.

In the matching process, we first identified all suppliers of firms listed on the North American market that did not have any GenAI announcements during the sample period. We then performed a logistic regression to estimate the likelihood of each supplier being included in our sample, using variables such as total assets, leverage, R&D intensity, market-to-book ratio, sales growth, and return on assets. Next, we calculated the propensity score for each supplier and applied nearest neighbor matching without replacement to find the closest match for each supplier within the same four-digit NAICS industry based on these propensity scores. If a match was not possible within the four-digit NAICS industry, we

broadened the search to higher-level industries until a match was found. The differences between our supplier sample and the matched firms are shown in Table A9 in Appendix EC.7.

Table 4 presents the event study results for the matched sample of suppliers around the announcement days of their counterparts in the original sample. The absence of a significant stock market reaction on the announcement day for these matched suppliers suggests that the significant ARs observed for suppliers of announcing firms are unlikely to be driven by sample composition in terms of observable supplier characteristics.

Table 4. Event study results for matched supplier sample.

	Day -1 (1)	Day 0 (2)	Day 1 (3)
Average abnormal return	-0.0009	0.0009	-0.0003
t-test (t-statistic)	(-0.9478)	(1.0460)	(-0.3313)
Median abnormal return	-0.0005	0.0005	0.0008
Wilcoxon signed rank test (Z-statistic)	(-0.5052)	(0.9868)	(0.1791)
Positive rate of the abnormal returns	0.4913	0.5146	0.5262
Binomial sign test (Z-statistic)	(-0.3966)	(0.6610)	(1.1898)
Observations	515	515	515

Notes: All tests are two-tailed. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Instrumental Variable Regression. To alleviate the potential concern over selection on unobservable characteristics, we employed instrumental variable regression to address the endogeneity in whether a supplier was chosen by an announcing firm. For this analysis, we combined the original set of suppliers with the matched suppliers and used a dummy variable, *Selected*, to indicate whether a supplier was selected by a GenAI announcing firm. We then used two-stage least squares (2SLS) to estimate the effect of *Selected* on the ARs of suppliers.

Our proposed instrument for *Selected* is the number of AI-related patents a supplier received between 2015 and 2019 (*AI Patent*), sourced from the Artificial Intelligence Patent Dataset in the USPTO database. Since the number of AI patents reflects a supplier's AI capabilities, it is expected to influence the supplier's likelihood of being selected by an announcing firm. However, because this information was publicly available well before the GenAI announcements, it is unlikely to directly cause *abnormal* stock price movements for the supplier. Therefore, *AI Patent* is a plausible instrumental variable.

Table 5 presents the regression results estimated by 2SLS. The coefficient of *Selected* in the second stage is significantly positive, indicating that GenAI announcements have a positive impact on the ARs of announcing firms' suppliers. The consistent finding suggests that our event study result cannot be attributed to selection on unobservable supplier characteristics either.

Table 5. Results of instrumental variable regression.

Variables	Whether the supplier is in our sample (<i>Selected</i>)	Abnormal return on day 0 (AR_0)
	First stage (1)	Second stage (2)
<i>Selected</i>		0.0061* (0.0033)
<i>Size</i>	−0.0040 (0.0066)	−0.0006* (0.0003)
<i>Age</i>	0.0028*** (0.0008)	−0.0001** (0.00004)
<i>MTB</i>	0.0022*** (0.0005)	−0.0000 (0.00002)
<i>Leverage</i>	0.0296 (0.0718)	0.0026 (0.0033)
<i>AI Patent</i>	0.0713*** (0.0063)	
Industry FE	No	Yes
Kleibergen-Paap rk Wald <i>F</i> statistic	23.407	—
Observations	1,030	1,030

Notes: The robust standard errors reported in parentheses are clustered at the supplier level. All continuous variables are winsorized at their first and 99th percentiles. *AI Patent* is log-transformed after adding one. The Kleibergen-Paap rk Wald *F* statistic of the weak identification test is larger than the critical value of the Stock-Yogo weak ID test at the 10% bias level, indicating that the instrument is fairly strong. MTB = market-to-book ratio; FE = fixed effect; AI = artificial intelligence. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

5.3.2 Robustness of Cross-Sectional Regression Analysis to Self-Selection. Our cross-sectional regression analysis in Section 5.2 may also suffer from the self-selection issue in the sense that the sample only includes suppliers of firms with GenAI announcements. This analysis

excluded suppliers of non-announcing firms, as one can argue that the ARs due to GenAI announcements are not well-defined for such suppliers. In light of this issue, we estimated a Heckman sample selection model to account for the potential self-selection in which suppliers are included in our regression analysis. To ensure that all supplier-level characteristics are properly defined, we conducted this analysis on the matched sample obtained in Section 5.3.1.

In the sample selection model, we continued to use *AI Patent* as an instrumental variable that appears in the first stage but not the second stage. While the inclusion of an instrumental variable in the first stage is not mandatory, it is often recommended to improve the identification of the sample selection model (Puhani, 2000). It should be noted that our results are similar with or without using this instrument, indicating that our findings do not hinge on the validity of this instrument.

Table 6 reports the results of the Heckman sample selection model, where Column 1 shows the first-stage results, while Columns 2 to 4 present the second-stage results using the same specifications as in Table 3. Note that the first stage remains the same regardless of the second-stage specification. Even after accounting for sample selection, the findings on the four supplier characteristics remain highly consistent with those in Table 3, suggesting that our results are robust to the potential sample selection bias.

Table 6. Cross-sectional regression results using the Heckman sample selection model.

Variables	Whether the supplier is in our sample (<i>Selected</i>)	Abnormal return on day 0 (AR_0)		
	First stage (1)	Second stage (2)	Second stage (3)	Second stage (4)
<i>Supplier R&D Intensity</i>		0.0158** (0.0072)	0.0170** (0.0072)	0.0196*** (0.0045)
<i>Supplier Sales Growth</i>		0.0120*** (0.0040)	0.0118** (0.0048)	0.0103** (0.0041)
<i>Supplier–Customer Distance</i>		−0.0010** (0.0004)	−0.0012** (0.0004)	−0.0011*** (0.0004)
<i>Supplier Industry Concentration</i>		0.0133** (0.0055)	0.0027 (0.0065)	0.0024 (0.0048)
<i>Product-Oriented Announcement</i>				0.0043** (0.0021)
<i>Rel Duration</i>		−0.0003 (0.0003)	−0.0003 (0.0003)	−0.0003* (0.0002)
<i>Announcement Intensity</i>		0.0008 (0.0008)	0.0016 (0.0030)	0.0001 (0.0012)
<i>Same Industry</i>		−0.0042 (0.0034)	−0.0049 (0.0042)	−0.0006 (0.0021)
<i>Size</i>	−0.0049 (0.0193)	−0.0009** (0.0005)	−0.0009 (0.0006)	−0.0014*** (0.0004)
<i>Age</i>	0.0060** (0.0025)	−0.0000 (0.0001)	−0.0001 (0.0001)	0.0000 (0.0001)
<i>MTB</i>	0.0079*** (0.0018)	−0.0000 (0.00002)	0.0000 (0.00002)	0.0000 (0.00002)
<i>Leverage</i>	0.0442 (0.2164)	0.0038 (0.0040)	0.0054 (0.0040)	0.0040 (0.0035)
<i>AI Patent</i>	0.2240*** (0.0219)			
<i>Inverse Mills Ratio (IMR)</i>		−0.0033 (0.0037)	−0.0039 (0.0046)	−0.0056 (0.0035)
Announcing firm FE	No	Yes	Yes	No
Industry FE	No	No	Yes	Yes
Observations	1,030	515	515	515

Notes: The robust standard errors reported in parentheses in the second stage are clustered at the announcing firm level. All continuous variables are winsorized at their first and 99th percentiles. R&D = Research and Development; MTB = market-to-book ratio; FE = fixed effect; AI = artificial intelligence. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

As a side note, neither the coefficient of the inverse Mills ratio nor the estimated two-stage correlation is insignificant, indicating that our estimation results are likely not biased by self-selection ([Hendricks et al., 2015](#)). We also ran a Heckman sample selection model in which *AI Patent* is included in both stages ([Table A10 in Appendix EC.8](#)), and the results are similar. Furthermore, the coefficient of *AI Patent* in the second stage is very close to zero and statistically insignificant, indicating that this variable may have no direct impact on ARs and hence likely satisfies the exclusion restriction.

5.3.3 Additional Robustness Checks. To ensure the robustness of our findings, we conducted several supplementary analyses, detailed in the [Appendices](#). (1) To address firm-level unobservables and temporal trends, we employed an event-based DID model with two-way fixed effects at the firm and time level, and the results remained consistent. (2) We examined cumulative ARs (CARs) over a period of up to 120 trading days following the announcement, and the results show that the cumulative effect is statistically significant and increases over time, suggesting a sustained impact on the market. (3) Replicating our analyses for traditional AI-related announcements, we found no significant ARs for suppliers, highlighting the distinct impact of GenAI announcements. (4) Using a 10-day window to exclude confounding events did not alter the main findings. (5) The results held after excluding announcements from firms that experience confounding events within a four-day window (days -2 to $+1$), those posted on or one day after major large language model (LLM) releases, and those posted on the same day or within four subsequent trading days of major government AI policies. (6) Findings remained consistent when isolating or controlling for GenAI infrastructure providers. (7) Excluding the top 5% of suppliers by total assets did not affect the main results. (8) Results were robust to the use of both Fama–French three-factor and five-factor models for AR estimation. [Table 7](#) summarizes all robustness checks.

Table 7. Summary of robustness tests.

Concern	Test	Finding	Location
Our findings are driven by the unique composition of suppliers	Propensity score matching	The matched firms do not experience a significant abnormal return	5.3.1
Our findings are driven by self-selection on unobservable supplier characteristics	Instrumental variable regression	Suppliers chosen by announcing firms have significantly higher abnormal returns than those not chosen	5.3.1
Self-selection in which suppliers are included in the cross-sectional analysis	Heckman sample selection model	Results remain consistent	5.3.2
Lack of control for time-invariant firm-level unobservables and temporal trends	DID analysis using matched suppliers without downstream GenAI announcements as controls	Downstream GenAI announcements improve the stock performance of suppliers	EC.9
Lack of long-term impact	Calculate cumulative abnormal return over extended periods	GenAI announcements generate a long-term positive impact on supplier stock performance	EC.10
Our findings are not unique to GenAI	Examining the impact of traditional AI announcements on suppliers	Traditional AI announcements do not generate significantly positive abnormal returns for suppliers	EC.11
Our findings are driven by the confounding events of suppliers	Identifying suppliers' events with a larger confounding event window	Results remain consistent	EC.12
Our findings are driven by confounding events of announcing firms, leading LLM releases, and government AI policies	Identifying and announcing firms' events, leading LLM releases and government AI policies	Results remain consistent	EC.13
Our findings are driven by GenAI infrastructure providers	Estimating effects separately for GenAI infrastructure providers and non-providers	Findings are similar for providers and non-providers	EC.14
Our findings are driven by a small subset of large suppliers	Excluding suppliers in the top fifth percentile of total assets.	Results remain consistent	EC.15
Our findings are sensitive to how we calculate abnormal returns	Calculate abnormal returns using the Fama–French three-factor and five-factor models	Results remain consistent	EC.16

LLM = large language model; DID = difference-in-differences; GenAI = generative artificial intelligence; AI = artificial intelligence.

6 Conclusion

GenAI is revolutionizing firm operations and supply chain management. Drawing on the relational perspective of disruptive innovation theory ([Kumaraswamy et al., 2018](#)), this article delves into the spillover effect of GenAI initiatives within the supply chain, particularly focusing on how downstream GenAI announcements influence the stock performance of suppliers. Through a comprehensive analysis of data on news, supply chain relationships, and daily stock prices, our findings reveal that GenAI announcements lead to a positive AR of 0.27% for suppliers on the announcement day. Additionally, we find that this spillover effect is more pronounced for suppliers with higher R&D intensity, strong sales growth, closer geographical proximity to their customers, operating in less competitive industries, and when GenAI announcements focus on product innovation. In a post-hoc analysis, we find evidence that GenAI announcements generate long-term positive stock market effects for suppliers, with ARs continuing to accumulate over an extended period.

Our findings align with the relational perspective of disruptive innovation, indicating that GenAI is a disruptive and transformative force that reconfigures relationships within supply chains. We observe significantly positive stock market reactions among suppliers, especially those that are more innovative or linked to product-oriented announcements. In contrast, traditional AI initiatives show no significant spillover effects. This divergence suggests that GenAI initiatives entail higher demands for supplier adaptation, collaboration, and technological alignment. These characteristics are consistent with GenAI's general-purpose

nature, its creative and generative capabilities, and its potential to reshape value creation across supply chain ecosystems. Thus, our results empirically substantiate the idea that GenAI constitutes a “bigger” or more systemic innovation compared to traditional AI.

6.1 Managerial Implications

Based on our empirical findings, we present key operational and strategic takeaways for supply chain stakeholders navigating the adoption of GenAI technologies. First, we reveal the positive spillover effect of downstream GenAI initiatives on suppliers’ short- and long-term stock market value, highlighting the significant influence that disruptive technologies like GenAI have on supply chain partners. We encourage managers to explore and adapt to the evolving GenAI-driven business ecosystem to remain competitive.

Second, given GenAI's transformative impact on supply chain dynamics, downstream firms should consider adapting their supplier relationships to better support technological integration. In particular, firms are advised to communicate their GenAI strategies and R&D roadmaps proactively and transparently to suppliers. Early and open communication enables suppliers to anticipate technological requirements and respond more effectively, fostering mutual trust and smoother collaborative innovation. This is especially critical in industries like consumer electronics, where product development depends heavily on supplier input ([Eisenhardt and Tabrizi, 1995](#)).

Third, managers of upstream firms must closely monitor and adapt to technological shifts among downstream customers. Suppliers with higher R&D intensity and strong sales growth are better positioned to benefit from GenAI-driven transformations. These firms should consider maintaining or increasing their innovation investments to align their capabilities with downstream demands. For example, Broadcom's investment in advanced semiconductor technologies to support GenAI infrastructure illustrates the strategic value of such alignment ([Reuters, 2024](#)). In addition, suppliers should consider establishing or strengthening their local presence near key downstream customers, especially in industries where collaboration and rapid innovation cycles are critical. For instance, technology component suppliers could establish regional offices or joint innovation centers to enhance collaborative innovation with downstream GenAI-adopting customers. Furthermore, suppliers in more competitive industries may need to adjust their product positioning and market segmentation to enhance their competitive advantage facing downstream disruptive innovation. For instance, AMD's launch of new AI processors in response to Nvidia's dominance exemplifies how firms can adapt their offerings to compete amid technological disruption ([Cherney and Bajwa, 2024](#)).

Finally, our findings indicate that suppliers experience significantly higher abnormal stock returns following product-oriented GenAI announcements compared to process-oriented ones. This underscores the need for suppliers to closely monitor customers’ R&D roadmaps and align their offerings accordingly, especially when they emphasize new AI-powered products or GenAI-enhanced features. For example, a hardware manufacturer serving smart-home device makers can develop GenAI-enabled hardware to enhance voice recognition and predictive automation rather than offering standard components. Such strategic adaptation to downstream product innovation signals strong innovation capability, promotes deeper technological integration, and may ultimately boost supplier visibility and market valuation. While process-oriented initiatives may not trigger significant short-term ARs, this does not imply they are less valuable; rather, they may yield long-term operational and financial

benefits that are not immediately priced in by the market. For instance, [Cheng and Nault \(2012\)](#) found that downstream firms' IT investments can improve upstream suppliers' productivity in the long run.

6.2 Limitations and Future Research

This research has certain limitations. First, we exclusively relied on major and commonly used news agencies to identify these GenAI announcements, which may have caused us to miss some relevant announcements. Future research could incorporate more comprehensive and diverse data sources, such as industry reports and social media posts, to capture a broader range of GenAI announcements. Second, our sample focuses on firms within the North American market and does not include those in emerging markets, which often have distinct social relations and political ties ([Lo et al., 2018](#)). These differences might influence how investors in suppliers react to technological innovations from downstream customers. Future studies could expand the geographic scope to include emerging markets, allowing researchers to explore potential heterogeneity in supplier reactions driven by variations in institutional contexts and investor behaviors. Lastly, while this study reveals GenAI's positive spillover effects on suppliers' ARs, the analysis primarily reflects investors' expectations at the time of the announcement and does not delve into the specific channels that drive these spillovers. Future research could examine the precise mechanisms, such as innovation spillovers or efficiency gains, through which disruptive innovations like GenAI generate long-term financial benefits for stakeholders in the supply chain.

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